Self-Supervised Learning for Chest X-ray Classification

1. Motivation

Deep learning models for chest X-ray classification need large numbers of labelled images, but labels are expensive because they require radiologists. Large collections of unlabelled radiographs, on the other hand, are plentiful in hospital archives.
Self-supervised learning (SSL) uses the unlabelled images to learn useful visual features first, and then needs only a small labelled set to adapt the model to a diagnostic task.

2. What is self-supervised learning?

In self-supervised learning the training signal is created from the data itself. A pretext task is designed so that solving it forces the network to learn meaningful representations. After pretraining, the pretext head is discarded and the encoder is reused for the real task, either by training a linear classifier on frozen features (linear probing) or by fine-tuning the whole network.

3. Contrastive methods

Contrastive methods such as SimCLR and MoCo create two augmented views of the same image and train the encoder to pull their embeddings together while pushing apart embeddings of different images. SimCLR relies on large batches so that each image has many negatives, and uses a temperature-scaled loss called NT-Xent. MoCo instead keeps a queue of past embeddings produced by a slowly updated momentum encoder, which allows many negatives with a small batch.

Non-contrastive methods such as BYOL and DINO remove the need for negatives. BYOL trains an online network to predict the output of a target network whose weights are an exponential moving average of the online weights.

4. Masked image modelling

Masked autoencoders (MAE) hide a large fraction of image patches, often 75 percent, and train the network to reconstruct the missing pixels from the visible patches. Because the encoder only processes visible patches, pretraining is efficient. Masked modelling is attractive for medical images because it does not depend on augmentations that might change the diagnosis.

5. Chest X-ray datasets

Several public datasets are used for pretraining and evaluation. NIH ChestX-ray14 contains 112,120 frontal-view images from 30,805 patients with 14 labelled thoracic findings. CheXpert contains 224,316 radiographs from 65,240 patients, with labels extracted from reports and an uncertainty label for ambiguous findings. MIMIC-CXR contains 377,110 images with free-text reports. The labels in these datasets are mostly produced by automatic text mining, so they contain noise.

6. Adapting augmentations to X-rays

Augmentations designed for natural images do not always suit radiographs. Strong colour jitter is meaningless on grayscale images, and horizontal flipping can swap the position of the heart, which matters clinically. Better choices include random resized crops with moderate scale, small rotations, brightness and contrast changes, and Gaussian blur. The augmentation strength should be validated, because overly aggressive cropping may remove the lesion that determines the label.

7. Evaluation protocol

Chest X-ray classification is usually a multi-label problem, since one image can show several findings. The standard metric is the area under the ROC curve (AUC) computed for each finding and then averaged. Because the classes are imbalanced, accuracy is a poor choice. Results should be reported with different amounts of labelled data, for example 1, 10 and 100 percent, to show the benefit of pretraining when labels are scarce.

To avoid leakage, images from the same patient must never appear in both the training and test splits.

8. Challenges and limitations

SSL pretraining is computationally expensive, and results can be sensitive to batch size, augmentation choices and training length. Domain shift between hospitals, scanners and patient populations can reduce performance. Label noise in the benchmark datasets limits how reliably small differences between methods can be measured. Finally, any model intended for clinical use needs external validation and attention to fairness across age, sex and ethnicity groups.